GEOMETRY-AWARE BIPLANAR CHEST CT RECONSTRUCTION WITH RESIDUAL LATENT DIFFUSION

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ABSTRACT

Reconstructing three-dimensional (3D) chest CT from biplanar X-ray projections requires inferring volumetric anatomy from limited views. We propose UPath-X2CT, a framework integrating geometry-aware feature fusion with residual latent diffusion. Calibrated projection matrices lift multiscale features onto a shared 3D grid, where adaptive fusion supports coarse latent prediction. Conditional diffusion models the latent residual, and decoded samples are averaged to obtain the final reconstruction. Evaluated on CT-derived projections from LIDC-IDRI, MSD Lung, and CT-RATE, UPath-X2CT outperforms five baselines in PSNR, SSIM, and MAE, with PSNR gains of 1.23–1.80 dB over DVG-Diffusion. Component and sampling analyses highlight the roles of geometric lifting and sample averaging. Code and pretrained model weights are available at https://github.com/hekaiwang/UPath-X2CT.

***Index Terms*—** Biplanar X-ray, chest CT reconstruction, geometric feature lifting, residual latent diffusion

1. INTRODUCTION

Biplanar chest computed tomography (CT) reconstruction aims to recover three-dimensional (3D) thoracic anatomy from anteroposterior (AP) and lateral (LAT) X-ray projections [1, 2]. These orthogonal views provide complementary structural cues, but superimpose anatomy along each projection direction. Volumetric reconstruction therefore requires cross-view geometric correspondence and learned anatomical priors. Fig. 1 illustrates the reconstruction setting.

X2CT-GAN [1] and PerX2CT [2] learn biplanar CT reconstruction through adversarial training and perspective-projection feature mapping, respectively. DiffuX2CT and DVG-Diffusion [3,4] explore conditional diffusion for X-ray-to-CT reconstruction, and EA-LDM [5] combines explicit latent alignment with patient-specific planning-CT priors. These approaches motivate combining geometry-aligned structural estimation with residual modeling in a compact latent space.

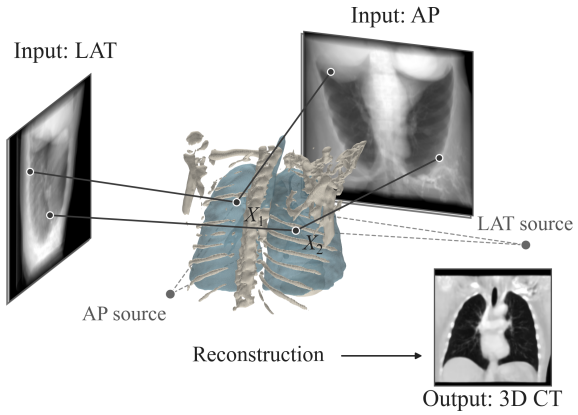


Fig. 1. Biplanar chest CT reconstruction. Calibrated AP and LAT projections provide complementary views of a shared 3D anatomical space.

We propose UPath-X2CT, a geometry-aware framework combining biplanar feature fusion with residual latent diffusion. A shared encoder and cross-view attention extract multiscale features, which calibrated projection matrices lift onto a shared 3D grid. Adaptive fusion and a global context branch support coarse CT latent prediction. Conditioned on the coarse estimate and fused features, diffusion models the residual to the target latent. At inference, sampled residuals are added to the coarse latent and decoded into CT volumes, whose average yields the final reconstruction.

The main contributions are:

- Geometry-aware multiscale lifting and adaptive fusion for spatially aligned biplanar feature integration.
- A coarse-to-fine framework combining deterministic latent prediction with residual diffusion.
- Consistent improvements in PSNR, SSIM, and MAE over five baselines on LIDC-IDRI, MSD Lung, and CT-RATE.

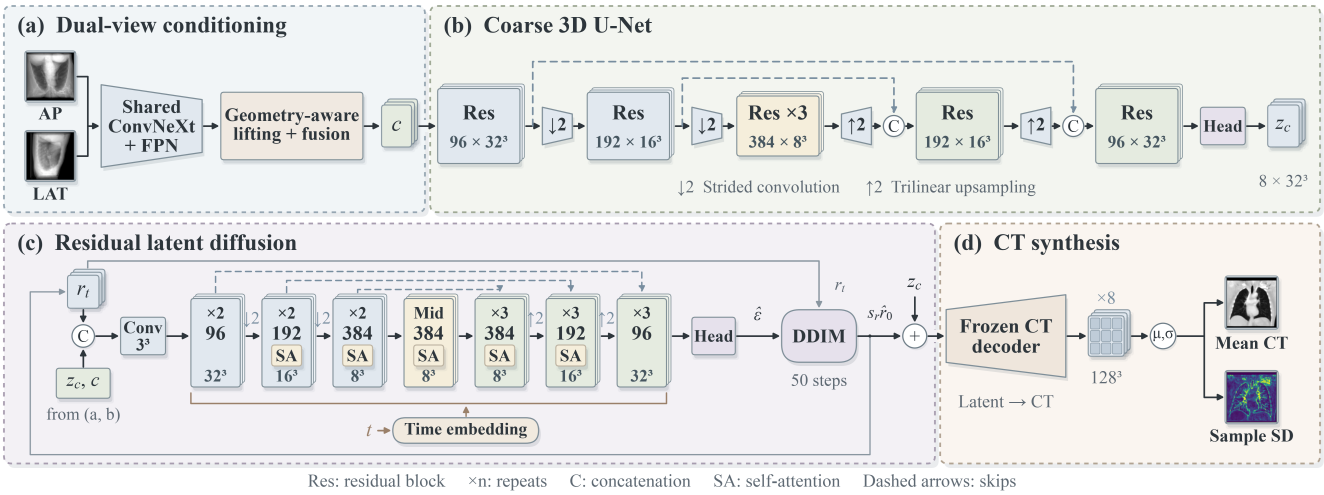


Fig. 2. Overview of UPath-X2CT. (a) Geometry-aware biplanar feature conditioning. (b) Coarse CT latent prediction using a 3D U-Net. (c) Conditional residual diffusion with DDIM sampling. (d) CT decoding and estimation of the mean reconstruction and voxelwise standard deviation ($K = 8$). Res: residual block; SA: self-attention; C: concatenation.

2. METHOD

2.1. Geometry-Aware Conditioning

Given calibrated AP and lateral projections (I_v, P_v), $v \in \{\mathrm{AP}, \mathrm{LAT}\}$, UPath-X2CT reconstructs the chest CT volume x. Following latent-space generative modeling [6], a separately trained 3D autoencoder maps normalized 128^3 CT volumes to an eight-channel latent $z = E(x)$ of spatial size 32^3, with decoder D recovering the CT volume.

As shown in Fig. 2(a), a shared ConvNeXt [7] with a feature pyramid network [8] extracts multiscale features, with cross-view attention at the two deepest levels. Features are lifted onto a physical 3D grid through calibrated projection and differentiable bilinear sampling:

$$f_v^\ell(\mathbf{q}) = m_v(\mathbf{q})\, \mathcal{B}\big(F_v^\ell, \pi(P_v \tilde{\mathbf{q}})\big)\,, \tag{1}$$

where $\mathbf{q}$ denotes a physical coordinate, $\tilde{\mathbf{q}}$ its homogeneous representation, ℓ the feature level, π perspective division, $\mathcal{B}$ bilinear sampling, and m_v detector visibility. Learned weights fuse features across visible views. View-specific and fused features, inter-view statistics, geometric encodings, and pooled global context are integrated into the 3D conditioning tensor c.

2.2. Coarse Reconstruction and Residual Diffusion

A 3D U-Net [9] with residual blocks predicts the coarse latent z_c from c, yielding $x_c = D(z_c)$. Its objective combines latent regression, CT intensity, multi-window, gradient, high-frequency, SSIM, and AP/LAT reprojection losses, alongside auxiliary body/lung and heteroscedastic supervision.

Table 1. Dataset composition and patient-level splits.

Dataset	Patients	CT volumes	Train/Val/Test	Evaluation
LIDC-IDRI	1,010	1,018	817/91/102	Internal hold-out
MSD Lung	63	63	0/0/63	External test
CT-RATE	425	425	0/0/425	External test

With the autoencoder and coarse network frozen, conditional diffusion models the normalized latent residual using Gaussian forward noising [10]:

$$\begin{aligned} r_0 &= (z - z_c) \oslash s_r, \\ r_t &= \sqrt{\bar{\alpha}_t}\, r_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \end{aligned} \tag{2}$$

where s_r is the channel-wise residual standard deviation, $\oslash$ denotes element-wise division, $\bar{\alpha}_t$ the cumulative noise coefficient, and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. A time-conditioned U-Net predicts noise from (r_t, z_c, c, t), trained with clipped-SNR-weighted noise prediction [11], smooth-L_1 residual regression, and decoded CT fidelity losses at lower noise levels.

At inference, DDIM sampling [12] generates K residual estimates $\hat{r}_0^{(k)}$. The final reconstruction is

$$\hat{x} = \frac{1}{K} \sum_{k=1}^{K} D\Big(z_c + s_r \odot \hat{r}_0^{(k)}\Big)\,, \tag{3}$$

where $\odot$ denotes element-wise multiplication. Voxelwise sample standard deviation summarizes reconstruction variability.

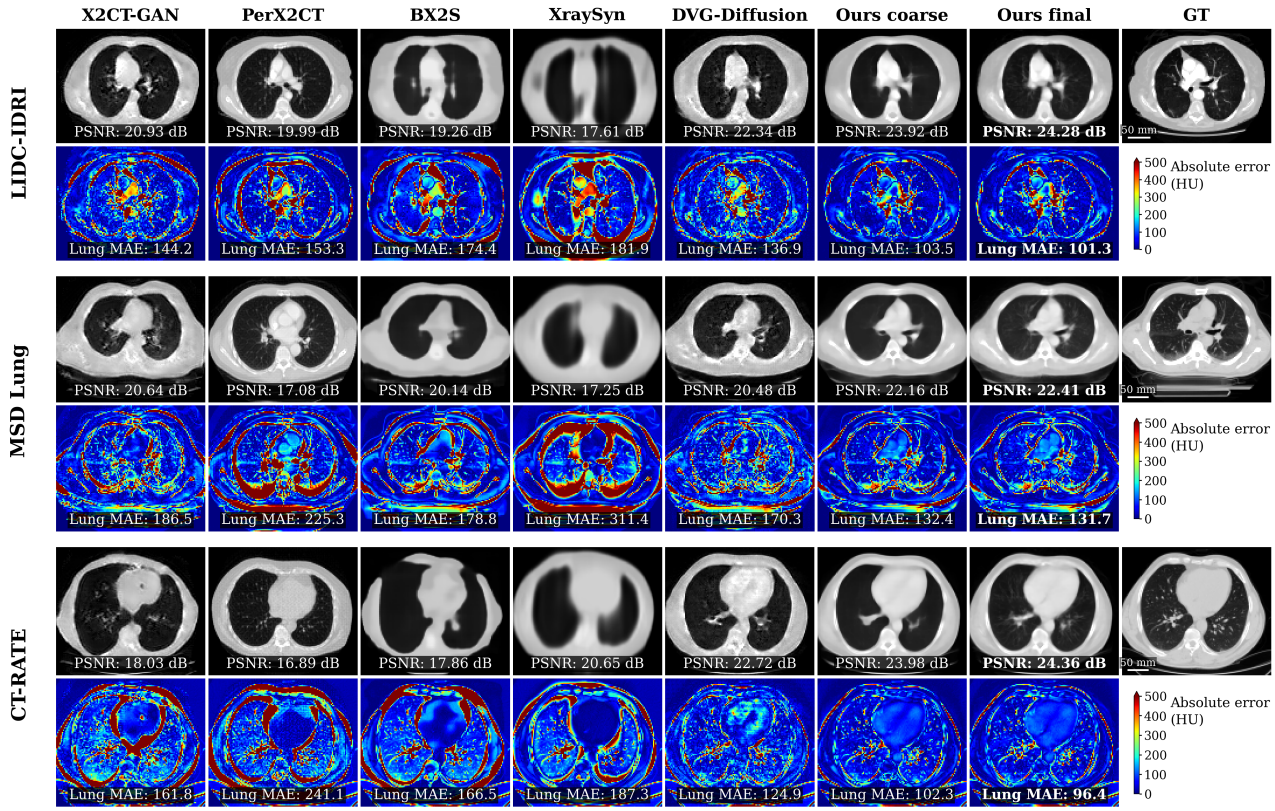


Fig. 3. Axial CT reconstructions and absolute error maps on LIDC-IDRI, MSD Lung, and CT-RATE (top to bottom). Error maps share a 0–500 HU scale; PSNR and lung MAE are overlaid on each reconstruction and error map, respectively. GT: ground truth.

Table 2. Reconstruction performance on three datasets. PSNR (dB), SSIM (%), and MAE (HU). Best and second-best results are bold and underlined, respectively.

Method	LIDC-IDRI					MSD Lung					CT-RATE				
	PSNR ↑	SSIM ↑	MAE ↓			PSNR ↑	SSIM ↑	MAE ↓			PSNR ↑	SSIM ↑	MAE ↓		
			Whole	Body	Lung			Whole	Body	Lung			Whole	Body	Lung
X2CT-GAN	24.05	72.67	69.49	131.67	163.80	18.33	53.47	166.62	334.62	174.38	17.48	47.19	196.79	426.36	195.12
PerX2CT	18.73	58.15	140.86	239.05	234.84	18.04	52.53	160.55	265.37	230.13	16.95	44.40	195.67	316.42	275.47
BX2S	21.75	66.18	94.47	158.35	187.65	21.54	64.10	101.87	164.20	162.09	21.15	59.65	109.96	178.01	194.58
XraySyn	19.39	59.20	138.96	228.13	215.20	18.47	51.51	171.58	259.78	228.24	18.57	48.27	163.49	244.44	243.89
DVG-Diffusion	24.55	74.18	65.33	127.36	146.47	23.47	68.91	79.19	133.64	150.13	22.91	64.47	85.64	145.60	171.05
Ours (coarse)	25.38	78.31	56.36	107.76	**113.39**	24.85	76.31	**62.63**	107.40	128.10	24.06	71.73	**70.04**	119.37	140.50
Ours	**26.35**	**81.45**	**50.00**	**96.14**	116.31	**24.98**	**76.43**	63.14	**105.24**	**125.06**	**24.14**	**71.84**	70.99	**117.82**	**139.61**

2.3. Implementation Details

ImageNet-pretrained ConvNeXt-Tiny processes 512×512 projections, with FPN levels P3–P5 producing a 96×32^3 conditioning tensor. Both 3D U-Nets use channel widths of 96, 192, and 384, with diffusion self-attention at 16^3 and 8^3.

The coarse and diffusion networks are trained sequentially using AdamW (effective batch size 4, learning rate 10^{-4}, weight decay 10^{-4}), with an encoder learning rate of 10^{-5}. Training uses 3,000 warm-up updates, cosine decay, BF16 precision, and exponential moving averages. Diffusion employs a cosine noise schedule [13] with 1,000 timesteps. The validation-selected checkpoint at 22,000 updates is evaluated with 50 DDIM steps, $\eta = 0$, and $K = 8$ samples.

Table 3. Component ablation. PSNR (dB), SSIM (%), and MAE (HU).

Geo.	Attn.	Adapt.	Diff.	PSNR ↑	SSIM ↑	MAE ↓
✓	✓			25.02	78.21	59.93
	✓	✓		22.94	70.73	77.71
✓		✓		25.28	78.18	<u>56.14</u>
✓			✓	<u>25.42</u>	78.23	57.16
✓	✓	✓		25.38	<u>78.31</u>	56.36
✓	✓	✓	✓	**26.35**	**81.45**	**50.00**

Table 4. Sampling ablation with 50 DDIM steps. PSNR (dB), SSIM (%), and MAE (HU).

K	PSNR ↑	SSIM ↑	MAE ↓		
			Whole	Body	Lung
1	25.40	77.48	57.56	113.04	134.87
2	25.92	79.52	53.61	104.25	125.06
4	<u>26.20</u>	<u>80.75</u>	<u>51.30</u>	<u>99.06</u>	<u>119.78</u>
8	**26.35**	**81.45**	**50.00**	**96.14**	**116.31**

3. EXPERIMENTS

3.1. Datasets and Preprocessing

LIDC-IDRI [14] is split into patient-disjoint training, validation, and test sets; MSD Lung [15] and CT-RATE [16] serve as external test cohorts (Table 1). CT volumes are resampled to thorax-centered 128^3 grids at 3-mm isotropic spacing, clipped to $[-1024, 1500]$ HU, and normalized to $[-1, 1]$. Calibrated cone-beam projections yield paired 512×512 AP/LAT DRRs. Reference body and lung masks are obtained by thresholding and morphological processing.

3.2. Baselines and Evaluation Metrics

We compare with X2CT-GAN [1], PerX2CT [2], HU-adapted BX2S [17] and XraySyn [18], and DVG-Diffusion [4], all trained and validated on LIDC-IDRI. XraySyn uses AP-only input; all other methods use both views. Metrics include PSNR (2,524-HU range), three-plane averaged SSIM, and whole-volume, body, and lung MAE (HU).

3.3. Results

UPath-X2CT outperforms all five baselines across three datasets (Table 2). Compared with DVG-Diffusion, it gains 1.23–1.80 dB in PSNR and 7.27–7.52 percentage points in SSIM, and reduces whole-volume and lung MAE by 17.11–23.47% and 16.70–20.59%, respectively. Relative to the coarse estimate, diffusion refinement consistently improves PSNR, SSIM, and body MAE on all three datasets.

Fig. 3 presents axial reconstructions and error maps. UPath-X2CT better preserves thoracic contours and lung boundaries, with refined intrapulmonary structures and the lowest lung MAE in all three examples.

3.4. Ablation Experiments

Both ablations use the 102-patient LIDC-IDRI test set. Table 3 evaluates geometric lifting (Geo.), cross-view attention (Attn.), adaptive fusion (Adapt.), and residual diffusion (Diff.). Removing Geo. causes the largest drop, reducing coarse PSNR from 25.38 to 22.94 dB. Adding Diff. to the full coarse model further raises PSNR by 0.97 dB, whereas Diff. without Attn. and Adapt. reaches only 25.42 dB.

With 50 DDIM steps (Table 4), increasing K from 1 to 8 raises PSNR by 0.95 dB and SSIM by 3.97 percentage points while consistently reducing all MAEs.

4. CONCLUSION

UPath-X2CT couples calibrated biplanar feature lifting with coarse-to-fine residual latent diffusion for chest CT reconstruction. On the internal LIDC-IDRI test set and the external MSD Lung and CT-RATE cohorts, it consistently achieves higher PSNR and SSIM and lower MAE than five baseline methods. Ablations confirm the contributions of geometry-aware conditioning, residual diffusion, and multi-sample averaging. These findings demonstrate the effectiveness of geometry-guided latent refinement for biplanar CT reconstruction, and future work will extend validation to clinical radiographs.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used public, de-identified data from LIDC-IDRI [14], MSD Lung [15], and CT-RATE [16]; therefore, no additional ethical approval was required.

6. ACKNOWLEDGMENTS

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ChatGPT (OpenAI) was used only for language polishing; all technical content was produced and verified by the authors.

The authors declare no competing financial or non-financial interests.

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